

BuildTrack: five phases from empty repo to live sites

This plan goes with the client PRD. For each phase it lists what we build, the requirements, the edge cases that break construction apps in real use, and the exit criteria a phase must pass before the next one starts.

How this maps to the PRD: PRD phase 1 is build phases 1–3, PRD phase 2 is build phase 4, and PRD phase 3 plus the pilot is build phase 5.

Flutter (Android · iOS · Web)

Firestore: Firestore · Auth · Storage · Functions · FCM · Hosting

2–3 developers

About 17–20 weeks

Phase 1 Foundation & platform 2–3 weeks
Phase 2 CEO app & project hub 4 weeks
Phase 3 Finance & approvals 3 weeks
Phase 4 Field apps & offline 4–5 weeks
Phase 5 Performance, bonus & go-live 4 weeks

These decisions apply across the whole codebase. Most of the edge cases in later phases are handled by getting these right once.

MONEY

- Store amounts as **integer paise** (`amountPaise: 1250000`). Never use doubles.
- Format with `intl en_IN` (₹12,50,000), with lakh/crore short forms on the dashboard.
- All sums happen server-side in Functions. The client never writes totals.

DATES & TIME

- Company timezone in `config/company.timezone` (default `Asia/Kolkata`).
- A "working day" is a string key `YYYY-MM-DD` computed in that timezone. It is used for DPR and attendance doc IDs.
- Moments are stored as `serverTimestamp()` plus `deviceTime`. Trust the server; keep the device value for audit.
- Scheduled Functions declare `timeZone` explicitly.

DERIVED DATA

- Triggers are at-least-once and may arrive out of order. **Recompute from source** (sum the category's approved expenses) instead of `increment()` deltas where the set is small. Where it isn't, use idempotent deltas keyed by `event.id`.
- A **nightly reconciliation job** recomputes every total and logs any drift it corrects.

CHANGE-TOLERANT DATA

- Every doc has `schemaVersion`, `createdAt/By`, `updatedAt/By` and `deleted` (soft delete only).
- Model parsing is tolerant: unknown keys are kept in `extra`, missing keys use defaults, and a bad field never crashes a screen.
- Lists and labels come from `config/*`, never from hard-coded enums (except system states like approval status).

ACCESS

- Custom claims hold **only** `role` and `companyId` (the claims limit is 1000 bytes, so project lists don't fit).
- Project access uses `projects/{pid}.memberIds[]`. Rules check `request.auth.uid in get(project).data.memberIds`, and child docs copy `memberIds` down to avoid extra

- `get()` reads. App Check is on for Firestore, Storage and Functions.

AUDIT

- Every create, update, approve or reject writes to `activity/` via a Function (who, what, before/after for money fields).
- Nothing is hard-deleted by users. Admins can archive.
- A daily managed Firestore export goes to a GCS bucket with 30-day retention. Point-in-time recovery is enabled in prod.

Phase 1

Foundation & platform

2–3 weeks

This phase produces no visible features. It builds the base every later screen depends on: auth, roles, config, the dynamic form engine, rules, CI and the design system. Figma sign-off by the CEO happens in parallel during week 1.

BUILD

- Two Firebase projects, `buildtrack-dev` and `buildtrack-prod`, set up through FlutterFire with flavors (`--dart-define=ENV`)
- Flutter app skeleton: Riverpod, `go_router` with role guards, frozen models, a responsive shell (sidebar on web and bottom nav on mobile), theme tokens and `en_IN` l10n scaffolding
- Auth: phone OTP on mobile, and email plus password or phone on web (web phone auth needs reCAPTCHA)
- Callable `admin.setUserRole` and `admin.inviteUser`, with a `users/{uid}` profile
- `ConfigRepository`, which streams `config/*` and caches it offline
- **Dynamic form engine:** renders `text | number | money | date | select | multiselect | photo | file | toggle` from field definitions, with validation
- Firestore and Storage rules with an emulator test suite
- CI with GitHub Actions: analyze, test, rules tests and Functions tests. Deploys to dev on merge; prod is deployed manually.
- Crash reporting: Crashlytics on mobile and Sentry on web (Crashlytics does not support web)

DATA CREATED

```
users/{uid}      role, name, phone, email, active,
                  fcmTokens[], lastSeenAt
config/company   name, timezone, currency, fyStartMonth
config/customFields {entity: [{id, key, label, type,
                             options[{id,label}], required,
                             showOnCard, order, archived}]}
config/lists     projectStatuses, projectTypes,
                  phaseTemplates, expenseCategories,
                  issuePriorities (each item has a stable id)
```

Edge cases

CASE	HANDLING	RISK
User deactivated while logged in	Set <code>active:false</code> , revoke refresh tokens (<code>revokeRefreshTokens</code>) and force a claim refresh. Rules also check a <code>disabledUsers/{uid}</code> marker so access ends immediately, without waiting up to an hour for the token to expire.	High
Role changed (supervisor promoted to manager)	Claims are updated server-side and the client listens to <code>users/{uid}.claimsVersion</code> . When it changes, the client calls <code>getIdToken(true)</code> and re-routes.	Med
Same phone number re-used by a new employee	An invite links a phone to a <code>users</code> doc. A phone number already tied to an inactive user needs the admin to confirm reassignment, and the old history stays with the old uid.	Med
OTP SMS not received or rate-limited	Resend timer, visible error text, and admin-set test numbers for demos. Configure the SMS region policy to India only to prevent SMS pumping costs.	Med
Custom field renamed, retyped or removed	<code>key</code> and <code>type</code> are immutable once used, and only <code>label</code> can change. Removing a field sets <code>archived:true</code> : values are kept but hidden. Select values store the option <code>id</code> , so renaming an option is safe.	High
A required custom field is added after records exist	Old records are not blocked. They show a "missing info" chip and the requirement applies only on the next edit.	Med

CASE	HANDLING	RISK
Old app version reads a newer doc shape	Tolerant parsing. A <code>config/app.minVersion</code> check shows a friendly "please update" screen on mobile.	Med
Someone accidentally writes to prod from dev	Separate projects and service accounts. The dev build shows a visible "DEV" banner, and prod deploys need a manual approval step in CI.	Med

Exit criteria

- Rules test suite passes: each role reads and writes only what it should (at least 40 cases)
- An admin can invite a user, assign a role, deactivate them, and access stops within 1 minute
- Adding a custom field in config renders it in a sample form on web and mobile with no deploy
- CEO has approved the Figma for the 4 drill-down levels

Phase 2

CEO app & project hub

4 weeks

The CEO's drill-down (portfolio → stage list → project overview → tabs), plus the admin web forms that feed it: projects, phases, daily targets and DPRs, issues and documents. Finance tabs show placeholders until phase 3.

BUILD: CEO

- **Portfolio home** reads a single `stats/portfolio` doc (1 read, instant)
- **Stage list:** a paginated query by `status` sorted by `healthRank`, with search on `nameLower` prefix and filters for manager and city
- **Project overview** reads the project doc (which carries its summary fields) and the latest 6 photos
- **Tabs:** Timeline (planned vs actual Gantt), Team, Daily progress (calendar and DPR detail), Issues, Documents & photos, Info (including custom fields)
- Global "Needs you" inbox shell (`approvals` where status is pending)

BUILD: ADMIN ENTRY (WEB)

- Project create/edit with the status lifecycle and custom fields

- Phases from a template, editable, drag to reorder, with a weight per phase
- Daily targets and DPR entry form, with multiple photo upload
- Issues and document upload with a type tag
- Functions: `onPhaseWrite` / `onDprWrite` → project `plannedPct`, `actualPct`, `health`; `onProjectWrite` → `stats/portfolio`
- Scheduled: daily health recalculation (planned % moves with the calendar even when nothing is written) and an issue escalation job

Health and progress formulas

```
plannedPct(project, today) = Σ phase.weight × clamp((today - plannedStart) / (plannedEnd - plannedStart))
actualPct                   = Σ phase.weight × phase.pct                    (phase.pct from DPR)
gap                         = plannedPct - actualPct
daysBehind                 = today - date on which plannedPct would equal actualPct
health = RED    if gap > 10 or daysBehind > 7 or budgetUsedPct - actualPct > 15
          AMBER if gap > 3  or daysBehind > 2
          GREEN otherwise
healthRank = RED 0 · AMBER 1 · GREEN 2 · NO_DATA 3    (thresholds live in config/health)
healthRank = RED 0 · AMBER 1 · GREEN 2 · NO_DATA 3    (used to sort "most at risk")
```

Phase weights default to equal and are normalised to add up to 1 on save.

Edge cases

CASE	HANDLING	RISK
Project put On Hold	The planned-progress clock pauses. <code>holdPeriods[]</code> are added to phase planned dates in the calculation, and health shows "On hold" instead of red.	High
Invalid status jump (Completed → Pipeline)	Allowed transitions come from <code>config/lists.projectStatuses[].next[]</code> and are enforced in a callable. Only admins can override, and the override requires a reason.	Med
Phase missing dates, zero-length, or end before start	Form validation. The calculation treats such a phase as excluded and the project shows "Timeline incomplete" instead of fake numbers.	Med
Pipeline project with no phases	Health is <code>NO_DATA</code> . Pipeline cards show value, probability and expected start instead of progress.	Med

CASE	HANDLING	RISK
Two DPRs for the same site and day	Doc ID <code>{pid}_{YYYY-MM-DD}</code> , so the second write edits the same report. Edits after submission keep a <code>revisions[]</code> trail.	High
DPR achieved quantity exceeds target, or is negative	Negative values are blocked. Values over 150% of target show a confirmation warning; they are allowed but flagged.	Med
Backdated DPR (entered 3 days late)	Allowed within <code>config.dprBackdateDays</code> (default 7). Its <code>submittedLate</code> flag counts in the phase 5 scoring.	Med
Completed project edited later	Completed projects are locked read-only. Reopening one is an admin action that is logged.	Med
Portfolio stats drift	Nightly reconciliation recomputes <code>stats/portfolio</code> from projects, and the CEO screen shows "updated hh:mm".	Med
Large photos and slow site uploads	Compress on the client to 1600px at quality 75, strip EXIF GPS where not needed, and use the Resize Images extension to create 400px thumbnails. Lists only load thumbnails.	Med
Firestore <code>in</code> or <code>array-contains-any</code> limits (30 values)	Never filter by long lists of project IDs. Query on a <code>memberIds</code> array-contains for the user, or read the denormalised stats doc.	Med
Flutter web slow first load for the CEO	CanvasKit with deferred imports for heavy tabs (Gantt, charts), and Hosting cache headers. Target first view under 3 seconds on 4G.	Med

Exit criteria

- Seed of 25 projects across every stage: portfolio numbers match a spreadsheet calculation exactly
- Health colours are correct for scripted scenarios: on track, slipping, behind, on hold, no data
- CEO usability test: finds "most behind ongoing project" in under 60 seconds with no help
- CEO role cannot edit any data (rules tests plus manual check)

Finance & approvals

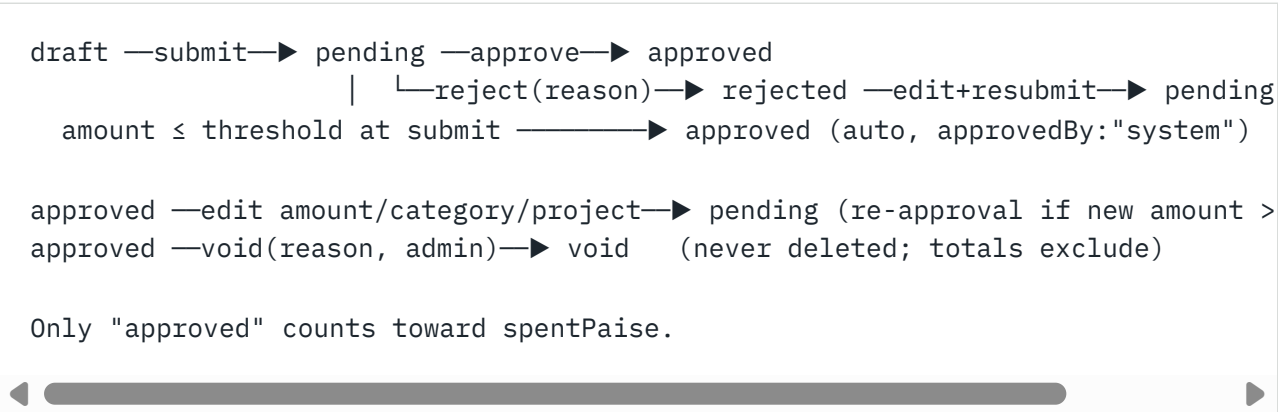
3 weeks

Budget against actual cost per category, expenses with bill photos, CEO approval above a threshold, alerts, the evening summary and exports. This is where mistakes cost real money, so it carries the most edge cases.

BUILD

- `projects/{pid}/budget/{categoryId}` with `plannedPaise` and `spentPaise` (spent is Function-only). Budget revisions keep history.
- `expenses/{id}` entry form: bill photo, custom fields, duplicate detection
- `onExpenseWrite` routes each expense to auto-approve or to a pending approval, recomputes the category and project totals, and writes to the activity log
- Callable `approvals.decide(id, decision, reason)`, which runs in a transaction
- CEO Money tab: category bars, burn rate, forecast at completion, and the expense list with bill viewer
- Alerts job for category 80% and 100% and for spend running ahead of progress. FCM plus in-app inbox.
- Evening summary push (config time, default 19:00)
- Exports: a callable builds XLSX or PDF in Functions, stores it in Storage, and returns a signed URL valid for 15 minutes

EXPENSE STATE MACHINE



Edge cases

CASE	HANDLING	RISK
Splitting one bill to stay under the threshold	A duplicate and split detector flags expenses with the same project, payee and category within 48 hours	High

CASE	HANDLING	RISK
	whose sum exceeds the threshold. They go to the CEO with a "possible split" label.	
Same bill entered twice	Hash of the bill photo, plus a soft match on payee, amount and date. The form warns before submit, and a flag shows in the approval list.	High
Approved expense edited afterwards	Edits to amount, category or project move it back to pending when the rules require. The old values stay in <code>revisions[]</code> and totals follow the current approved state only.	High
Two approvers act at once, or a double tap	<code>decide()</code> runs in a transaction that requires <code>status == pending</code> . The second call gets "already decided by X" and the UI disables the button while the call is in flight.	High
Threshold changed while expenses are pending	Each expense stores <code>thresholdAtSubmit</code> . A change applies only to new submissions, and existing pending items stay pending.	Med
Expense category deleted from config	Categories can only be archived. Archived categories keep their totals and are hidden from new entry.	Med
Expense moved to another project	Both projects' totals are recomputed, because a <code>recompute from source</code> handles this automatically.	Med
Expense on a completed or on-hold project	Completed: blocked unless an admin reopens it. On hold: allowed, with a warning.	Med
Budget revised mid-project	<code>budgetRevisions[]</code> keeps the date, old and new amounts, reason and user. The CEO sees the original and the current budget.	Med
No budget set for a category that has spend	Shown as "Unbudgeted spend" in red. It is never divided by zero.	Med
Forecast at completion with 0% progress	The forecast is hidden until <code>actualPct ≥ 10%</code> and shows "Not enough progress yet".	Med
Expense date in a closed month	<code>config.finance.lockedUntil</code> . Entries before that date need an admin override with a reason.	Med
Alert spam (80% alert every night)	Each alert has a key such as <code>{pid}_{cat}_80</code> and fires once. It re-arms only if spend drops below the	Med

CASE	HANDLING	RISK
	level and crosses it again.	
Big export times out	The export Function runs with 540s and 1GB, and streams rows. Anything over 50k rows is split by month.	Med
Bill photo leaks by URL	Storage rules require project membership, and bill links are signed and short-lived. Download URLs are never stored in public docs.	High

Exit criteria

- A scripted run of 1,000 expenses with random edits, moves, rejections and voids gives totals equal to an independent recompute (zero drift)
- Concurrency test: 10 parallel `decide()` calls on one approval leave exactly 1 decision
- Duplicate and split detection catches all seeded cases in the test set
- CEO approves 5 pending expenses from the phone in under 60 seconds

Phase 4

Field apps & offline

4–5 weeks

Managers and supervisors take over data entry from the office. The hard part is **offline reliability**: sites have weak or no signal, phones are low-end and shared, and people close apps mid-upload.

BUILD: MANAGER

- "My projects" list with the same overview as the CEO, plus edit rights
- Update phases and progress, set daily targets (copy from yesterday), log expenses and raise or assign issues
- Approve staff leave, and review and return DPRs

BUILD: SUPERVISOR (MOBILE ONLY)

- Today screen: targets, then attendance, then DPR, then submit
- **Labour attendance**: crew list, "all present" in one tap, then exceptions (absent, half-day, overtime hours)

- **DPR in under 2 minutes:** quantities prefilled, labour count taken from attendance, at least one photo required, issues inline
- Workers register (labourers without a login): name, trade, daily wage, ID photo optional
- Reminders: FCM at 18:00, then an 8 pm escalation to the manager, then the CEO's "missing reports" list

Offline architecture

- Firestore offline persistence is on (mobile default), with the cache size set to 100 MB. Writes queue locally and sync automatically.
- **Firestore Storage does not queue uploads offline**, so we build a local **upload queue** (drift/SQLite table: `localPath, storagePath, docRef, field, attempts, status`), drained by `workmanager` on connectivity and at app start.
- A DPR doc is written immediately with `photos: [{id, status:"pending"}]`. The queue uploads each file and updates it to `{status:"uploaded", path}`, and the CEO view shows "photo syncing" for pending ones.
- Photos are copied into app storage before queuing, so they survive the gallery or cache being cleared.
- A sync status bar shows "3 items waiting to sync". The supervisor can see and retry them; items are never silently lost.
- Logout is blocked, with a clear message, while the queue has unsynced items.

Edge cases

CASE	HANDLING	RISK
Supervisor offline all day, syncs next morning	The DPR doc ID uses the working day it was recorded for , not the sync time. <code>deviceTime</code> is kept and the server marks <code>syncedLate</code> , but it is not penalised when device time and the record date agree.	High
Phone clock set wrong or tampered	Compare <code>deviceTime</code> with <code>serverTimestamp</code> at sync. If the gap is over 10 minutes and the device was online at the time, flag the entry for manager review.	Med
Two people edit the same DPR offline	Per-target fields merge (<code>items</code> . <code>{targetId}.achieved</code> as map keys, not arrays), so most edits don't collide. If the same field is edited by both, last write wins and the loser is recorded in <code>revisions[]</code> .	Med

CASE	HANDLING	RISK
Labourer marked present at two sites the same day	An attendance write trigger checks <code>workerDay/{workerId}_{date}</code> . A conflict flags both entries for the managers, and the second entry counts as a half-day until resolved.	High
App killed mid-upload or photo corrupted	The queue retries with backoff (up to 10 attempts), then marks the item failed and visible with a "retake photo" action.	Med
Low storage on a cheap phone	Compress before queuing, and warn when free space is under 200 MB. Local copies are deleted once an upload is confirmed.	Med
Shared phone between two supervisors	Logout clears the local cache (<code>clearPersistence</code>) only after the queue is empty. The PIN/OTP login switches user.	Med
Supervisor assigned to multiple sites	A site picker appears on the Today screen, and the last-used site is remembered.	Med
Supervisor removed from a project while offline	Queued writes are rejected by rules at sync. The app shows "You no longer have access to Site X" with the unsent data, and offers to export it to the manager so nothing is silently dropped.	High
Holidays and Sundays	<code>config/calendar</code> holds weekly offs and holidays per project. No DPR is expected and there are no reminders or missing-report alerts on those days.	Med
Night shift that crosses midnight	The working day is set by the shift start. The supervisor can choose "yesterday" as the report date within 12 hours.	Med
Stale FCM tokens and multiple devices	Tokens are stored as an array with <code>lastUsed</code> . Tokens that fail with <code>registration-token-not-registered</code> are removed, and tokens older than 60 days are pruned.	Med
Reminder sent to someone on approved leave	Reminder jobs skip users with approved leave for that date and send to the backup supervisor if one is set.	Med

Exit criteria

- Airplane-mode test: a full day of attendance, DPR and 10 photos offline. After reconnecting, all of it syncs, with nothing lost or duplicated.
- Kill the app during upload 5 times: every photo eventually arrives
- DPR is completed in under 2 minutes by a real supervisor on a low-end Android phone (2 GB RAM)
- Double attendance conflict is detected in all seeded cases

Phase 5

Performance, bonus & go-live

4 weeks

Staff check-in, monthly scoring, the bonus run with CEO approval, then hardening and a two-site pilot before full rollout. Bonus mistakes damage trust with employees, so every number must be explainable.

BUILD

- Staff check-in and check-out: GPS inside the project geofence, plus a selfie
- Manager monthly rating: 1–5 stars with an optional comment, due by the 3rd of the next month
- Scheduled job on the 1st at 02:00 computes `scores/{YYYY-MM}_{personId}` with a full breakdown and the inputs used
- `bonusRuns/{YYYY-MM}` draft, then CEO review (edit any row with a reason), then Approve, which locks it. Export to payroll XLSX.
- Employee view: own score, breakdown and tips. Never other people's.
- Settings: weights (must sum to 100), score bands to bonus %, eligibility rules

SCORE FORMULA

```
attendance = presentDays / expectedWorkingDays      (approved leave excluded from
targets    =  $\sum$  achieved /  $\sum$  target (crew DPRs) | tasks on time / tasks due
quality    = 1 - min(1, reworkIssues / max(1, tasksDone) × k)
rating     = (stars - 1) / 4
score      = 100 ×  $\sum$  weighti × factori          (factor with no data → weight redistrib
bonus      = band(score).pct × monthlyWage × (eligibleDays / monthDays)
```

Edge cases

CASE	HANDLING	RISK
Joined or left mid-month	The bonus is pro-rated by eligible days. Anyone below <code>minEligibleDays</code> (default 10) is shown as "not eligible" with the reason.	High
Moved between projects mid-month	Factors are computed per project assignment period and weighted by days. The breakdown shows each project.	Med
No targets assigned or no rating given	The factor is marked "no data" and its weight is spread proportionally over the others, so a missing rating doesn't count as a zero. The CEO sees a "manager rating missing" warning.	High
Attendance or DPR edited after the bonus run	An approved run is locked. Edits that affect a locked month create an <code>adjustments/</code> entry that appears in the next month's run.	High
Weights changed mid-month	A run uses the weights snapshot taken at compute time, stored in <code>bonusRuns.settingsSnapshot</code> , so the run can always be explained later.	Med
GPS spoofing or poor accuracy	Reject <code>isMocked</code> on Android. Readings with accuracy worse than 100 m ask for a retry, and 3 failures allow a check-in flagged for manager confirmation.	High
Forgot to check out	A job at <code>config.autoCheckoutTime</code> closes the record at shift end, flagged. Hours are not credited beyond the shift end.	Med
Location or camera permission denied	An explanation screen with a button to open settings, and a fallback of manager-confirmed attendance.	Med
Labourer daily wage vs staff monthly wage	<code>monthlyWage = dailyWage × presentDays</code> for labour. Bands apply the same way and the payroll export shows both.	Med
Manager biased ratings	The rating is capped at 20% weight by default. The CEO sees each manager's rating distribution (all 5 stars is flagged).	Med
Rounding differences in payroll	Bonus is computed in paise and rounded to the nearest ₹10 (configurable) only in the final column. The draft total always equals the sum of the rows.	Med

CASE	HANDLING	RISK
Selfie privacy	Selfies are visible to the manager and admin only and deleted after 90 days by a lifecycle rule. Consent text is shown at first check-in.	Med

Hardening & go-live checklist (last 2 weeks) —

- Load test: 50 sites and 500 users for 30 days of seeded data. The CEO home screen loads in under 2 seconds.
- Firestore read cost review: no unbounded listeners on the CEO dashboard, and every list is paginated
- Budget alerts set on the GCP billing account
- Security pass: rules audit, App Check enforced, no secrets in the client, Storage paths reviewed
- Backup restore drill: restore yesterday's export into dev and confirm the data
- Pilot on 2–3 sites for 2 weeks, with a daily 10-minute feedback call with the supervisors
- Data migration: import existing projects and budgets from the client's Excel through a validated import screen with dry run and error report
- Training: a single one-page guide per role, plus 3 short videos in the local language
- Play Store internal track, then production; TestFlight, then App Store
- Support: an in-app "Report a problem" that attaches logs and a screenshot

Exit criteria (go-live)

- A bonus run for pilot staff matches a hand calculation for every person, and every number can be traced to its inputs
- No critical or high bugs open. Crash-free sessions are 99.5% or higher during the pilot.
- Pilot supervisors submit DPRs on at least 90% of working days without office help
- CEO signs off after using it daily through the pilot

Across all phases

Testing, quality and team process

LAYER	WHAT WE TEST	TOOLING
Business logic	Health, forecast, totals, score and bonus formulas, and date and working-day helpers (timezone and DST-free boundaries)	Dart <code>test</code> , Jest for Functions
Security rules	A matrix of every role × every collection × read/write, including cross-project access attempts	<code>@firebase/rules-unit-testing</code> on the Emulator
Functions	Triggers are idempotent (the same event run twice gives the same result) and handle out-of-order events and concurrency	Emulator plus Jest
UI	Dynamic form engine, key screens, role routing and empty, error and loading states	Widget and golden tests
End-to-end	DPR offline flow, expense to approval to totals, and bonus run	<code>integration_test</code> on real devices (one low-end Android)
Seed and simulation	A script generates projects, people and months of activity for demos and load tests	Node script against the Emulator or dev

EVERY SCREEN MUST HAVE

- Loading, empty, error and offline states designed, not left as defaults
- Large touch targets (at least 48px), outdoor-readable contrast, and text that stays readable at 130% system font size
- All labels in l10n files, ready for Hindi or Marathi later

DEFINITION OF DONE (PER FEATURE)

- Rules and tests written, edge cases from this plan covered
- Works on web, Android and iOS, offline where relevant
- Activity log entries created for data changes
- Demoed with seed data and reviewed against the PRD

Client inputs needed before phase 3

Expense approval threshold and approvers, budget categories (their Excel), and project stages. These are open questions 1–3 in the PRD. Phase 1 and 2 can start without them because all of these are config values.

BuildTrack build plan · v1 · 26 Sep 2026 · pairs with the client PRD v0.9